



Environmental Management Accounting: Awareness and Utilization Among Maltese Accountants



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Abstract: Increasing pressure on relevant stakeholders to shoulder environmental responsibility has compelled businesses to minimize their environmental impact through the provision of, say, improved accounting information. For the accounting profession to effectively contribute to this shift, practitioners should demonstrate both awareness and practical application of environmental management accounting (EMA). This study investigated the level of awareness and utilization of EMA among Maltese Certified Public Accountants (CPAs or accountants), while also examining the influence of job experience, environmental training, size of firm, and business sector. A quantitative research approach was adopted and data was collected through a self-administered online questionnaire distributed to Maltese accountants. Descriptive statistics were used to analyze frequencies of response, while the Chi-square and Kruskal-Wallis tests were applied to assess relationships and test hypotheses. Consistent with existing literature, the findings revealed that both EMA awareness and utilization were low to moderate. They also indicated that accountants who had been trained in EMA while occupying entry-level or executive roles in their organizations exhibited a slightly higher EMA awareness than others. On the other hand, those employed within the larger firms, particularly the Big Four, displayed a slightly higher EMA use than others, indicating differences across firm sizes and sectors. Despite the ongoing limitations, the presence of certain awareness and applications highlighted the existence of a real potential for gradual improvement. Over time, the increased EMA awareness and adoption could enhance environmental practices, and, ultimately, support the country's long-term environmental objectives, thus contributing to progress toward achieving carbon neutrality by 2050.

Keywords: Environmental management accounting; Environmental accounting; Environmental awareness; Environmental training; Accounting profession; Accounting practice; Certified public accountants; Malta

1. Introduction

Accounting has lately expanded its scope to support businesses in reducing their environmental impact through environmental management (Alkisher, 2018). Environmental management accounting (EMA), which is a subset of Environmental Accounting (EA), emerges as the system that performs this role by identifying, collecting, and analyzing environmentally related information (Jasch, 2003). This is particularly important given that traditional overhead allocation methods often produce distorted results, as they fail to adequately capture environmental costs and impacts (Krutilla & Krause, 2011).

EMA remains an evolving field of study since it could not be understood in a straightforward manner, partly due to the absence of a generally agreed upon definition. The limitation has led to differing perspectives among scholars, resulting in ongoing debates about its scope and implementation (Burritt et al., 2002). Early studies conducted by Bartolomeo et al. (2000), the United Nations Division for Sustainable Development (2001), Burritt et al. (2002) and Jasch (2003) explained that EMA was the internal system within the field of EA. It incorporated

both physical and monetary internal data, respectively known as Physical EMA (PEMA) and Monetary EMA (MEMA). The key distinction between these two subsets lies in their approaches of measurement. PEMA is unit-oriented as it reports environmental information in physical units, while MEMA is money-oriented and translates physical environmental data into monetary environmental costs. International Federation of Accountants (2005) provided common examples for each subset. Physical data includes the use, flow, and destinies of energy, water, and materials (including wastes) as well as consumption of resources and emissions related to water, energy, air, and waste. Monetary data capture environment-related costs, earnings, and savings.

More recently, the use of EMA has been extended to include external use, suggesting that it should be incorporated into external reporting. For instance, Agustia et al. (2019) and Johnstone (2020) contended that EMA had an external element, as stakeholders increasingly evaluated a company's environmental and financial performance when making decisions. Providing transparency in environmental performance, therefore, benefits both society and the firm's accountability. Ncube et al. (2023) commented that EMA contributed to broader reporting practices, such as integrated reporting, which examined how an organization's strategy, governance, performance, and prospects created value over short and long term. Such information could subsequently be used for external communication, to enhance transparency and stakeholders' trust.

2. Literature Review

2.1 Environmental Management Accounting Adoption

Traditional accounting systems often fail to adequately capture environmentally related costs. Consequently, several authors like Burritt et al. (2002), International Federation of Accountants (1998), Vărzaru et al. (2022), and Wahyuni (2009) advocated for the adoption of more advanced accounting tools. Common EMA techniques include Activity-Based Costing (ABC), Cost of Quality (CoQ), Life Cycle Costing (LCC), Input–Output (I–O) analysis, Materials Flow Analysis (MFA), and the Sustainability Balanced Scorecard (SBSC). Among these, ABC, CoQ, LCC, I–O, and MFA were primarily applied to environmental cost analysis, whereas the SBSC was widely used to monitor and evaluate environmental performance.

2.1.1 Activity-based costing

According to Hansen et al. (2021), ABC involves identifying activities, assigning costs to those activities, and allocating them to products based on consumption of resources. As such, ABC functions as a decision-support tool centered on how resources are utilized across activities. From an EMA perspective, Antić et al. (2021) extended this application by incorporating environmental performance indicators, such as pollution levels, recycling of resources, social impact, and remediation costs into costing practices, thereby enabling firms to account for environmental considerations when pricing products.

2.1.2 Cost of quality

An environmentally oriented CoQ framework focuses on identifying waste and improving process efficiency (Cokins, 2006). It categorizes environmental costs into prevention, inspection, internal failure, and external failure.

2.1.3 Life cycle costing

Environmental impacts may arise at multiple stages of a product's life cycle, requiring a broader costing perspective. LCC addresses these impacts by evaluating environmental burdens associated with products and processes over their entire life span (Chen et al., 2010). The approach provides an overall view of environmental costs, including energy and material usage as well as waste generation (United States Environmental Protection Agency, 1995), thereby supporting more informed decision making.

2.1.4 Input–output analysis

Input–output (I–O) analysis traces the environmental impacts associated with consumption patterns (Munksgaard et al., 2005). It distinguishes between inputs, representing the resources required for production, and outputs, representing the goods and services produced (Miller & Blair, 2022), thereby enabling the assessment of environmental impacts across economic sectors.

2.1.5 Material flow analysis

MFA evaluates the efficiency of resource utility by tracking flows of materials, water, and energy within a system. These flows are classified as inputs and outputs, including production, waste, and emissions. By defining system boundaries at the organizational level such as cost centers, processes, or products, MFA provides insights into environmental performance that are not captured by conventional accounting systems (United Nations Division for Sustainable Development, 2001).

2.1.6 Sustainability balanced scorecard

In 1992, Robert Kaplan and David Norton introduced the balanced scorecard to assist entities in meeting their strategic objectives. Nonetheless, it did not focus on environmental concerns. Eventually, the SBSC was developed, to allow businesses to identify environmental and social goals of a business (Elijido-Ten & Tjan, 2014). The SBSC focuses on five perspectives: “Social”, “Environmental”, “Learning and Growth”, “Internal Process” and “Economic” are integrated upon the vision of business, and all of which have their respective objectives, measures, targets, and initiatives. This strategic approach allows users to design their environmental perspective in line with the actual and desired objectives of the business. Consequently, it enables efficient allocation of resources for optimization of environmental performance (Deegan, 2003). Therefore, every business will have to analyze its customized and distinctive environmental goals.

2.1.7 Other environmental management accounting practices

Given that EMA is often criticized as costly and a potential burden, particularly discouraging for certain firm sizes (Borg, 2020; Wilmshurst & Frost, 2001), some authors proposed simpler and practical alternatives. Cernikova & Malíková (2015) and Jasch (2006) suggested integrating environmental aspects into sub-ledgers within accounting systems. For example, usage of resources such as materials, gas, and electricity may be recorded as expenditure, while emission-related charges can be classified under taxes and fees. Similarly, Phan et al. (2017) recommended embedding environmental data within accounting software to improve efficiency and enhance accountants’ familiarity with EMA concepts. Another approach emphasized assigning responsibility for environmental decision making. Cernikova & Malíková (2015) and Jasch (2006) highlighted the importance of designated employees, while Lodhia (2003) argued that responsibility should be shared among stakeholders, accountants, and environmental experts. Leadership support is critical to ensure employees understand and manage environmental costs effectively (Darnall et al., 2008; Kokubu & Nashioka, 2005). A further alternative is adopting a “re-thinking” approach to improve resource efficiency and e-waste management. For instance, using multifunctional devices instead of multiple machines could reduce waste. In addition, applying an e-waste hierarchy, prioritizing “reduce”, “reuse”, and “recycle” over “incinerate” and “landfill”, could enhance environmental performance (Gunaratne, 2015).

2.2 Challenges to Environmental Management Accounting Adoption: The Issue of Low Awareness

Empirical evidence across diverse geographical contexts pointed to a persistent and systemic challenge in EMA. This issue was initially highlighted by Burritt et al. (2002), who identified a significant gap between the theoretical advancement of EMA and its practical application. This disconnect was largely attributed to limited professional awareness, the absence of standardised frameworks, and inadequate training opportunities within the accounting profession.

This challenge was consistently reflected in both early and more recent empirical studies. Foundational research conducted by Bartolomeo et al. (2000) in Europe, Wilmshurst & Frost (2001) in Australia, Lodhia (2003) in Fiji, and Setthasakko (2010) in Thailand emphasized limited awareness of EMA concepts, minimal involvement of accountants in environmental management initiatives, and a general absence of EMA-related training. Collectively, these studies suggested that EMA had historically occupied a peripheral position within mainstream accounting practices. Similarly, more recent evidence has indicated that this issue remained unresolved. For instance, Qian et al. (2015) found that EMA development in China remained at an early stage, with implementation hindered by limited environmental awareness, insufficient organisational capabilities, and inconsistent regulatory support. These findings highlight the continuing need for technical expertise and formalised training to facilitate the integration of EMA into organisational systems. Similar patterns have been identified in Nigeria (Olalekan & Jumoke, 2017), South Africa (Nyahuna & Doorasamy, 2022), Bangladesh (Islam & Rahman, 2022), and Malta (Psaila et al., 2024). Taken together, these findings indicated that low awareness was not a context-specific issue but a pervasive global barrier. This exacerbates ongoing uncertainty among accountants regarding their role in environmental cost identification, measurement, and integration into strategic decision-making processes. The persistent gap underscores the requisites for enhanced professional education, clearer institutional guidance, and stronger integration of EMA within accounting curricula and practice. This is essential as accountants are considered “gamechangers” in advancing sustainability practices (Gibassier & Alcouffe, 2018),

2.3 Factors Influencing Environmental Management Accounting Awareness and Adoption

2.3.1 Environmental training

Effective application of EMA depends on proper employee training (Thaller et al., 2023), especially in emerging areas with limited awareness (Albelda, 2011; Bartolomeo et al., 2000; Burritt et al., 2002; International Federation of Accountants, 1998; Jasch, 2003). Professional programmes such as the Association of Chartered Certified Accountants (ACCA), the Chartered Institute of Management Accountants (CIMA), and university courses have

incorporated EMA concepts to build foundational knowledge. Apart from this, workplace training remains essential to foster environmental awareness and capabilities (Hoai et al., 2023; Jackson et al., 2023). Common methods include seminars, webinars, conferences, and on-the-job training (Gunarathne & Alahakoon, 2016; Jalaludin et al., 2011; Nguyen, 2020; Oladinrin & Ojo, 2022). Evidence suggested that such training improved EMA knowledge (Chathurangani & Madhusanka, 2019; Egan & Tweedie, 2018; Smit & Dikgwatlhe, 2015; Wilmshurst & Frost, 2001), while a lack of training limits accountants' effectiveness in environmental management (Qian et al., 2015). In this light, the following hypotheses have been developed:

H₀: Training does not affect accountants' awareness of EMA.

H₁: Training does affect accountants' awareness of EMA.

2.3.2 Experience of individuals

Individual experience, shaped by knowledge and professional attributes, plays a critical role in EMA adoption (Thaller et al., 2023). Prior studies highlighted the need for accountants to apply practical experience and adapt their competencies to improve environmental data quality and decision making (Birkin, 1996; Rogers, 2003). Research also indicated that senior management demonstrated higher EMA awareness and application (Amir et al., 2020; Latan et al., 2018), as they tended to be more responsible for making environmental-related decisions. On the other hand, younger accountants may exhibit stronger theoretical understanding due to increased academic exposure and initiatives of professional training (Jackson et al., 2023). Hence, the following hypotheses have been developed:

H₀: Job position does not affect accountants' awareness of EMA.

H₁: Job position does affect accountants' awareness of EMA.

2.3.3 Business size

Prior research generally suggested that business size played a significant role in the adoption of sophisticated accounting practices, with larger firms more likely to implement advanced systems (Bjørnenak, 1997; Cadez & Guilding, 2008; Chenhall, 2003; Haldma & Lääts, 2002). This relationship is commonly attributed to more availability of resources, which enables larger organizations to invest in and experiment with new accounting techniques (Innes & Mitchell, 1995). Extending this argument to EMA practices, studies such as Christ & Burritt (2013), Ferreira et al. (2010), and Nguyen (2020) similarly found that adoption is more prevalent among large firms, partly due to their capacity to absorb higher implementation costs as well as increased exposure to public scrutiny and public image.

Nonetheless, this perspective is not universally supported. Several studies such as Lima Ribeiro & Aibar-Guzman (2010), Phan et al. (2017), and Taufiq et al. (2017) reported an insignificant relationship between business size and EMA adoption, arguing that the fundamental purpose of EMA, to monitor and control environmental information, was relevant across organizations regardless of their sizes. This challenged the assumption that the availability of resources alone drove adoption and suggested that other factors, such as managerial awareness or organizational priorities, might play a more decisive role. Therefore, the relationship between business size and EMA adoption remains inconclusive, indicating a need for further investigation within specific contexts. Therefore, the following hypotheses have been developed:

H₀: Business size does not affect utilization of EMA.

H₁: Business size does affect utilization of EMA.

2.3.4 Business sector

Prior research indicated that the business sector might also influence EMA adoption. Frost and Wilmshurst & Frost (2001) suggested that firms in retail settings adopted environmental management practices that differed from those in extractive or chemical industries, while arguing that organizations operating in environmentally sensitive sectors are more likely to implement EMA. This view is supported by Ferreira et al. (2010). Similarly, Mokhtar et al. (2016) found that EMA implementation was more prevalent in environmentally sensitive industries, such as chemicals, energy, oil, and other sectors subject to greater environmental regulation and stakeholder scrutiny (Johnstone, 2020).

Qian et al. (2015) observed greater engagement with EMA among manufacturing organizations than in the public sector. They reflected the need to monitor and control environmental costs to improve efficiency. Christ & Burritt (2013) and Jamil et al. (2015) similarly emphasized the focus of manufacturing firms on productivity and cost reduction, positioning EMA as a practical tool in achieving these objectives. In this context, Yung et al. (2011) noted that techniques such as ABC and LCC were particularly suitable. These findings suggested that sectoral differences in EMA adoption were closely tied to operational priorities, especially requirements of cost control and efficiency. This suggests the following hypotheses:

H₀: Business sector does not affect utilization of EMA.

H₁: Business sector does affect utilization of EMA.

2.4 Research Gap in the Maltese Context

While previous international studies identified several factors that might influence EMA awareness and adoption, including environmental training, professional experience, business size, and business sector, limited evidence exists regarding whether these relationships apply within the Maltese context. Existing Maltese studies examined EMA from different perspectives. To name a few examples, Bugelli (2011) investigated the relationship among EMA, environmental performance indicators, and financial performance, while Callus (2014) examined accountants' involvement in energy-related investment decisions and environmental initiatives. Similarly, Borg (2020) explored EMA implementation through a Maltese organisational case study, and Sacco (2020) examined environmental indicators in relation to EMA practices.

These studies did not empirically examine the factors influencing EMA awareness and utilisation among Maltese Certified Public Accountants (CPAs). Consequently, it remains unclear to which extent characteristics such as environmental training, professional position, organisational size, and business sector affect CPAs to understand and apply EMA practices. Addressing this gap is particularly relevant given the increasing sustainability requirements imposed by the European Union, including the Corporate Sustainability Reporting Directive (2022), and Malta's position as a small island economy. Therefore, this study contributes to existing EMA literature by examining the determinants of EMA awareness and utilisation among Maltese CPAs.

3. Methodology

3.1 Research Tool

To assess Maltese accountants' awareness and use of EMA, a large sample quantitative approach was deemed most suitable (Saunders et al., 2019). The survey was divided into three sections and contained 26 questions. Section 1 collected information on the participant's personal and professional background, focusing primarily on gender, age, years of experience as an accountant, and the workplace settings of their current job. Sections 2 and 3 contained questions that assessed the understanding and use of EMA. Most questions utilised statements on a Likert scale ranging from 1 to 5, where 1 represents the lowest score and 5 represents the highest score.

3.2 Sample Population

As of 14 November 2023, there were 3,622 CPAs registered in Malta, as listed by the Malta Accountancy Board (Accountancy Board, n.d.) according to the requirements of the Accountancy Profession Act, Cap. 281 (1979) (Chapter 281 of the Laws of Malta). The required sample size of 348 respondents was calculated using a 95% confidence level and a 5% margin of error. Data were collected between 14 November 2023 and 10 January 2024; during which 369 completed responses were obtained, representing 10.1% of the population. The achieved margin of error was $\pm 4.84\%$, indicating improved precision over the initial target.

3.3 Data Analysis

Quantitative data were collected through an online and self-administered questionnaire, to ensure all participants answered identical questions without interviewer's influence. The data were analysed using IBM SPSS Statistics (Version 29). Descriptive statistics were used to analyse percentages of response, while non-parametric statistical tests, namely the Kruskal–Wallis test and the Chi-square (χ^2) test, were employed to examine relationships between variables. Statistical significance was assessed at the 0.05 level; the H_0 was accepted when $p > 0.05$, indicating no significant relationship, whereas the H_1 was accepted when $p < 0.05$, indicating a statistically significant relationship.

3.4 Reliability and Validity

The questionnaire was developed following a review of the EMA literature to capture respondents' awareness and use of EMA within their organisations. Demographic and organisational questions were adapted from recognised external classifications and statistical frameworks to ensure consistency, comparability, and reliability in measuring factors that may influence perceptions and EMA adoption. The questionnaire was pilot tested for clarity and validity. Where applicable, multi-item Likert scales achieved Cronbach's Alpha coefficients above 0.8, confirming good internal consistency and reliability.

4. Results

4.1 Profiles of Participants

Table 1 illustrates the demographic composition of the participants, who were almost evenly divided between

males (49.3%) and females (50.4%), with most of the population (40.9%) falling in the age range of 31–40. Besides, the number of years practising as a qualified accountant was uniformly shared across all selections, except for those who have been practising for over 10 years as they made up a smaller percentage (20.6%).

Table 1. Demography of respondents

Variables	Attribute	Frequency (N)	Percentage (%)
Gender	Male	182	49.3%
	Female	186	50.4%
	Others	1	0.3%
Age	21–30	137	31.7%
	31–40	151	40.9%
	41–50	60	16.3%
	51–60	18	4.9%
	61 and over	3	0.8%
Number of years practicing	Up to two years	99	26.8%
	More than 2 years up to 5 years	102	27.6%
	More than 5 years up to 10 years	92	24.9%
	More than 10 years	76	20.6%

4.2 Factors Influencing Environmental Management Accounting Awareness and Utilization

Based on the relevant literature review and prior studies, this research adopted four principal factors that might influence the level of awareness and utilization of EMA among participants. Table 2 presents the variables alongside the distribution of respondents according to their professional profiles. The findings indicated an almost equal likelihood of having received training (48.8%) versus not having received training (51.2%). Most respondents occupied senior-level roles (47.7%) were most likely employed in medium-sized firms (33.6%) and tended to either work in the industry (30.6%), small to medium practices (29.3%), or one of the Big Four accounting firms (22.8%).

Table 2. Profiles of respondents under study

Variable	Attribute	Frequency (N)	Percentage (%)
Received environmental training	Yes	180	48.8%
	No	189	51.2%
Job position	Entry-level	16	4.3%
	Mid-level	100	27.1%
	Senior-level	176	47.7%
	Executive-level	77	20.9%
Business size of current employment	Micro (<10 employees)	44	11.9%
	Small (10–49 employees)	72	19.5%
	Medium (50–249 employees)	124	33.6%
	Large (≥250 employees)	129	35%
Business sector of current employment	Big four	84	22.8%
	Small-Medium Practitioner (SMP)	108	29.3%
	Professional Accountant in Business (PAIB)	113	30.6%
	Public sector	41	11.1%
	Sole practitioner	18	4.9%
	Others	5	1.3%

4.3 Environmental Management Accounting Awareness among Maltese Accountants

When Maltese accountants had to rate their levels of awareness on a 5-point Likert scale, with 1 implying “Not aware at all” and 5 implying “Extremely aware” on topics relating to EMA, a medium level of awareness was established. EA had the highest mean score ($\bar{x} = 2.47$), followed by EMA ($\bar{x} = 2.28$) and its two subsets, MEMA ($\bar{x} = 2.23$) and PEMA ($\bar{x} = 2.12$). These results implied that overall speaking Maltese accountants had limited familiarity with EA concepts, with relatively greater awareness of broader frameworks than of more specific EMA practices.

When grouping awareness of EMA categories by environmental training (Yes/No) and job position (Entry/Mid/Senior/Executive), the Kruskal–Wallis tests revealed significant relationships in both cases ($p < 0.05$)

for all EMA categories (i.e., EA, EMA, PEMA, and MEMA). These findings led to the rejection of the H_0 , indicating that both environmental training and job position significantly influenced the level of awareness among Maltese CPAs. Moreover, higher mean scores were observed for all EMA categories among CPAs who received environmental training while occupying either an entry-level or an executive position. The two variables, i.e., Training and Job Position, did influence the level of EMA awareness.

The study also found that a significant number of participants faced challenges in accurately distinguishing physical and monetary environmentally related information. For instance, the neutral mean score for use, flows, and rates of environmental expenses ($\bar{x} = 2.93$) suggested that Maltese accountants were not fully aware of these primary physical indicators. Similarly, the environmental expenditure, assets, and liabilities ($\bar{x} = 3.32$) indicated that respondents were not fully aware that these represented monetary data.

Moreover, accountants were asked to rate on a 5-point Likert scale, where 1 indicated “Not relevant at all” and 5 indicated “Extremely relevant”, regarding the perceived relevance of the six sophisticated tools identified in Section 1. The SBSC registered the highest mean score ($\bar{x} = 3.61$), followed by CoQ ($\bar{x} = 3.52$), I-O ($\bar{x} = 3.35$), LCC ($\bar{x} = 3.33$), MFA ($\bar{x} = 3.32$), and ABC ($\bar{x} = 3.29$). These emphasized that accountants had a 50:50 chance of having sufficient knowledge regarding the effectiveness and application of EMA tools. One possible reason was accountants had a 50:50 chance of receiving training. When accountants were asked whether organizations needed to adopt sophisticated systems for identifying, managing, controlling, and accounting for environmental aspects of their operations, 77.8% responded “Yes”. This indicated clear recognition of the merit for more robust systems to support effective EMA implementation, even if understanding of EMA tools and practices remained limited.

4.4 Level of Environmental Management Accounting Adoption among Maltese Accountants in Their Current Employment

In Section 1, multiple studies highlighted that EMA was traditionally recommended for internal use and could nowadays be applied for external purposes. When accountants were asked to rate the frequency that their organization measured internal environmental performance, a mean score of $\bar{x} = 2.8$ was recorded. On the other hand, measuring the environmental performance of other organizations recorded a lower mean score of $\bar{x} = 2.58$. These results indicated a medium level of adoption of these EMA-related practices at their current employment. When this was categorized by business size (Micro/Small/Medium/Large) and business sector (Big 4/SMP/PAIB/Public Sector/Sole Practitioner/Others), the Kruskal–Wallis test demonstrated that the p -values for both variables (internal and external performance) were below the significance level of 0.05. Thus, the H_0 was rejected, thus indicating significant difference in the frequencies businesses tracked internal and external environmental performance. Moreover, the mean scores suggested that as business size increased, the frequency of measurement increased: the highest mean value was recorded in the Big Four. This indicated that larger firms and the Big Four tended to track internal and external environmental performance more frequently than smaller accounting firms.

When accountants were asked to rate the extent to which the six tools identified in Section 1 were currently adopted within their organisations (1 indicates “Never Adopted” and 5 indicates “Largely Adopted”), the overall utilisation was found to be relatively low. Among them, ABC was the most utilised tool ($\bar{x} = 2.47$), followed by CoQ ($\bar{x} = 2.41$), I-O ($\bar{x} = 2.31$), SBSC ($\bar{x} = 2.18$), LCC ($\bar{x} = 2.17$), and MFA ($\bar{x} = 2.05$). The results suggested that Maltese businesses were making only limited use of advanced management accounting tools in their day-to-day operations; they might even be further from effectively implementing such systems for EMA purposes. Furthermore, the results of the Chi-square test supported the acceptance of the H_0 when each tool was analysed across business sizes and sectors, indicating no statistically significant differences in adoption levels across organisational categories. This finding was reinforced by the consistently low to moderate mean scores observed across all business sizes and sectors.

In contrast, business size and sector appeared to influence the adoption of broader environmental management practices. This relationship was first observed in response to whether environmental-related tasks were allocated to a specific function within the organisation. When analysed by business size, respondents from large firms reported the highest proportion of “Yes” responses (45.7%), followed by medium-sized firms (34.7%), small firms (25%), and micro firms (13.6%). A similar pattern emerged across business sectors, with respondents from Big Four firms reporting the highest affirmative responses (58%), followed by PAIBs (34.1%), SMPs (29.6%), the public sector (25.7%), sole practitioners (11.1%), and others (0%). The Chi-square test supported the H_1 , as the p -value fell below the 0.05 threshold, indicating statistically significant differences across business size and sector.

This trend was further reinforced by responses to whether organisations employed personnel specifically responsible for environmental management. Consistent with previous findings, large firms again recorded the highest proportion of affirmative responses (66.7%), followed by medium-sized firms (31.5%), small firms (26.4%), and micro firms (13.6%). When examined by sector, Big Four firms had the highest level of dedicated environmental personnel (72.6%), followed by the public sector (46.3%), PAIBs (33.6%), SMPs (26.9%), sole practitioners (11.1%), and other sectors (20%). The Chi-square test of the organization-personnel relationship once

again accepted the H_1 , confirming that the adoption of such practices varied significantly by business size and sector.

4.5 Solutions for Maximizing the Use of Environmental Management Accounting

A concluding question was posed to all respondents to ascertain the most effective initiative for enhancing the utilization of EMA. The statement that received the highest frequency score was the recommendation to offer additional training, accounting for 37.9% of the responses. This finding was logical, given that more than half of the Maltese CPAs surveyed had never undergone any EMA training beforehand. This was followed by “Setting up monetary rewards for companies” at 20.1%, “Increased collaboration between accountants and environmental experts” at 19.8%, “Encouraging environmental audits” at 14.4%, and “Collaborative partnerships with environmental organizations” at 7.9%.

5. Discussion

The discussion presented in Section 5 is linked to the literature reviewed in Section 2 and the results obtained in Section 4. The findings of the present study were placed within a broader international context, thus contributing to a better understanding of EMA adoption and practices in Malta.

5.1 Environmental Management Accounting Awareness among Maltese Certified Public Accountants

The findings of this study indicated a persistently low level of EMA awareness among Maltese CPAs, as reflected in the relatively low mean scores across all measured EMA dimensions (i.e., EA, EMA, MEMA, and PEMA). This suggested that despite the growing global emphasis on sustainability and environmental reporting, EMA remained insufficiently understood at both conceptual and practical level within the Maltese accountancy profession. Respondents demonstrated limited ability to differentiate between physical and monetary environmental information, thus highlighting a lack of conceptual clarity in interpreting environmental data within accounting systems. Such ambiguity undermined the effective translation of environmental information into useful output for accounting decisions.

In addition, the findings revealed limited awareness of the applicability of advanced management accounting tools within an environmental context. Although overall recognition of such tools was low, a substantial proportion of respondents (77.8%) acknowledged the need for more sophisticated systems to manage and report environmental information. This pointed to a paradox in which the importance of environmental data was recognised, yet the technical knowledge required to operationalise EMA in practice remained underdeveloped. This raised questions regarding the extent to which current education and professional training adequately emphasised the practical tools necessary for effective EMA implementation.

These results align with existing literature that highlighted a persistent gap between the theoretical development of EMA and its practical application (Burritt et al., 2002). Empirical studies from various regions including Europe (Bartolomeo et al., 2000), Australia (Wilmshurst & Frost, 2001), Fiji (Lodhia, 2003), Thailand (Setthasakko, 2010), China (Qian et al., 2015), Nigeria (Olalekan & Jumoke, 2017), South Africa (Nyahuna & Doorasamy, 2022), Bangladesh (Islam & Rahman, 2022), and Malta (Psaila et al., 2024) consistently reported low EMA awareness and limited practitioners’ engagement. Despite growing regulatory and stakeholder pressure, EMA remains underutilized. In Malta, although CPAs have the potential to drive organizational sustainability, this potential is yet to be fully realized. These findings highlight the need for stronger institutional support through enhanced education, professional training, and organizational system.

5.1.1 Environmental training related to business aspects

Environmental training emerges as a significant determinant of EMA awareness among Maltese CPAs. The results indicated that 48.8% of respondents had received some form of environment-related training in their current employment, while a slightly higher proportion (52.8%) reported no such exposure. This suggested an uneven professional development across the profession. Respondents who had received training consistently reported higher mean scores across all EMA categories, thus indicating a positive association between training and EMA awareness. This relationship was further supported by the results of the Kruskal–Wallis test, which revealed a statistically significant difference ($p < 0.05$) between trained and untrained respondents. These findings reinforced the argument that structured professional training played a critical role in enhancing accountants’ understanding of EMA concepts and applications. Professional training serves as a key mechanism through which abstract sustainability principles are translated into practical accounting competencies. The results are consistent with prior studies by Duong et al. (2022), Hoai et al. (2023), and Lodhia (2003), all of which emphasised the importance of education and professional development in facilitating EMA adoption.

However, the relatively limited exposure to environmental training suggested that existing initiatives might be

insufficient in both scope and depth. This shortfall likely contributed to the observed disconnect between conceptual awareness and practical application. Accordingly, there is an apparent need for comprehensive and targeted training programmes that move beyond basic awareness and emphasise hands-on application of EMA tools. Innovative and interactive learning approaches may be particularly effective in equipping accountants with the skills required to integrate environmental information into management accounting systems. Strengthening such initiatives would not only enhance individual competencies but also support wider organisational adoption of environmentally responsible practices.

5.1.2 Job position

The findings further indicated that job position played a statistically significant role in shaping EMA awareness among Maltese CPAs. Specifically, individuals at entry-level and executive roles reported higher levels of awareness compared to those occupying intermediate positions. This suggested an uneven distribution of EMA knowledge across organisational hierarchies. Higher awareness among entry-level professionals may reflect recent developments in accounting education, where sustainability and EMA concepts are increasingly incorporated into university curricula and professional qualifications. Consequently, newer entrants to the profession may possess more current knowledge in this area. This is further supported by the Kruskal-Wallis test, which yielded a p -value below the 0.05 significance threshold.

At the executive level, increased awareness was likely associated with strategic responsibilities, including decision making related to environmental performance, regulatory compliance, and reputational risk. Individuals in senior roles may therefore have stronger incentives to engage with EMA concepts and environmental information (Amir et al., 2020; Latan et al., 2018). In contrast, lower awareness among intermediate-level professionals may signal gaps in continuous professional development. This group often occupies operational roles with limited exposure to strategic initiatives or advanced training opportunities, hence potentially constraining their engagement with EMA practices. The findings highlighted the importance of reinforcing EMA knowledge throughout the entire career trajectory rather than solely emphasizing it at entry or senior levels.

5.2 Environmental Management Accounting Practices in Maltese Organizations

The findings of this study indicated a generally low level of adoption of EMA practices among Maltese businesses. This is evidenced by limited monitoring of environmental performance, partial use of environmentally oriented management tools, and the absence of formal systems for the separate identification and disclosure of environmental costs. Collectively, these results suggested that EMA was not yet embedded within internal management processes and remained peripheral to core accounting practices.

A further concern was related to the lack of specialized EA expertise within organizations. The findings showed that environmental responsibilities were typically assigned to general accountants, who might not possess the necessary technical knowledge to effectively implement EMA. This not only limited the quality of environmental information produced but also constrained its usefulness for decision-making purposes. In addition, the adoption of specific EMA tools remained limited. While ABC appeared to be used to some extent, more advanced techniques were rarely applied, indicating a narrow and underdeveloped application of EMA methodologies within Maltese businesses.

These findings stood in contrast to normative literature, which emphasized the importance of integrating EMA into internal decision-making processes to improve cost control, resource efficiency, and environmental performance. Prior research (e.g., Burritt et al., 2002; International Federation of Accountants, 1998; Vărzaru et al., 2022; Wahyuni, 2009) advocated systematic identification of environmental costs, use of specialized tools, and involvement of trained personnel as key components of effective EMA implementation. The limited adoption observed in this study therefore suggested a gap between recommended practices and organizational realities. However, the results are consistent with earlier Malta-focused studies (Psaila et al., 2024), which similarly reported low levels of EMA adoption. This continuity over time indicated that progress in this area had been slow, despite increasing international pressure on enhanced sustainability reporting. A plausible explanation might lie in the relatively small size of the Maltese economy, limited regulatory enforcement, and a lack of institutional incentives for organizations to invest in sophisticated EA systems.

5.2.1 Business size

The relationship between business size and the adoption of environmental management accounting practices (EMAPs) was widely examined in prior literature, with most studies concluding that larger organizations were more likely to implement sophisticated accounting systems, including EMA (Bjørnenak, 1997; Cadez & Guilding, 2008; Chenhall, 2003; Ferreira et al., 2010; Haldma & Lääts, 2002; Innes & Mitchell, 1995). This is typically attributed to more availability of resources, increased stakeholder pressure, and higher levels of organizational complexity.

The findings of this study largely supported this perspective, as mean scores for EMAP adoption increased with

firm size. Larger Maltese organizations appeared more likely to implement environmentally-oriented accounting practices and utilize more advanced management accounting tools. This suggested that resource capacity and organizational structure played an important role in facilitating EMA adoption.

However, the results revealed that these differences were not consistently statistically significant across all variables. Notably, only the SBSC demonstrated a significant relationship with firm size. This indicated that while larger firms tended to adopt more sophisticated practices, the overall gap between small and large organizations might be less pronounced than in prior research. This finding added nuance to the existing literature by suggesting that firm size alone might not be a sufficient determinant of EMA adoption in smaller economies such as Malta. Other factors such as regulatory pressure, characteristics of the industry, and organizational priorities may play equally important, if not greater, roles. Consequently, future research should adopt a more multidimensional approach when examining the drivers of EMA implementation.

5.2.2 Business sector

The results of this study indicated that business sector was an important factor influencing the adoption of EMAPs, although its impact varied across different practices. Consistent with prior research such as those conducted by Christ & Burritt (2013) and Ferreira et al. (2010), the findings suggested that sector-specific characteristics such as regulatory exposure, stakeholders' expectations, and operational complexity shaped the extent to which organizations engaged with EA.

In the Maltese context, Big Four firms demonstrate the highest level of EMAP adoption across all measured areas. This is likely attributable to their global exposure, access to advanced resources, and strong emphasis on professional training and best practices. Moreover, these firms play an essential role in shaping the competencies of future accountants through early career development, thereby contributing to the diffusion of EMA knowledge. In contrast, lower levels of adoption are observed in the public sector, among sole practitioners, and within PAIB roles. This disparity suggests that EMAP diffusion is uneven across sectors, potentially reflecting differences in institutional pressures and resource availability. For instance, smaller practices and public sector entities may face budgetary constraints or lack the strategic incentives required to prioritize EMA implementation.

Meanwhile, the present findings supported the view that not all environmental practices were sector dependent. Simpler initiatives such as the use of environmentally friendly workplace tools appear to be adopted more broadly, regardless of the type of industry. This suggested that while complex EMA systems might require sector-specific drivers, more basic environmental practices could be diffused more easily across organizational contexts.

5.3 The Way Forward

The findings of this study provided important insights into the most effective strategies for promoting EMA adoption among Maltese CPAs. The results indicated that the provision of additional training was perceived as the most effective measure, reflecting a clear gap of skills within the current profession. This is particularly significant given that a substantial proportion of respondents reported no prior exposure to environmental training in a business context. This highlighted the critical role of professional bodies, educational institutions, and businesses in facilitating EMA adoption through structured learning initiatives. Training should not be limited to theoretical concepts but should emphasize practical application, enabling accountants to integrate environmental information into routine decision-making processes. In this regard, more innovative and interactive training approaches may enhance engagement and improve learning outcomes.

In addition to training, respondents also identified monetary incentives and increased collaboration between accountants and environmental experts as important enablers of EMA adoption. Financial incentives may encourage businesses to invest in the necessary systems and expertise, while interdisciplinary collaboration could help bridge the knowledge gap between accounting and environmental management. Conversely, strategies such as environmental auditing and partnerships with environmental organizations were perceived as less effective. This may reflect concerns regarding their cost, complexity, or limited familiarity within the Maltese context. However, these tools may still play a complementary role in supporting transparency and accountability, particularly as regulatory requirements continue to evolve.

Overall, the findings suggested that advancing EMA in Malta would require a coordinated effort involving regulatory support, professional training, and organizational commitment. Considering the increasing European Union requirements for sustainability reporting, failure to address these gaps may place Maltese organizations at a disadvantage. Strengthening EMA capabilities is therefore not only an environmental imperative but also a strategic necessity for ensuring long-term competitiveness and compliance.

6. Conclusions

The current study achieved its objectives through an online questionnaire distributed to Maltese CPAs, hence yielding a sufficiently large sample to generalize findings. Statistical analysis revealed that overall awareness of

EMA among Maltese CPAs was limited, with many struggling to distinguish between its key components. Statistical tests concluded that environmental training and the CPA's job position played crucial roles in determining the level of EMA awareness among participants, while business size and business sector influenced the level of adoption of EMA practices. Despite these limitations, the presence of certain awareness and application highlighted the potential for gradual improvement. Over time, the increased awareness and adoption of EMA could strengthen environmental practices, and ultimately, support the long-term environmental objectives of the country, including progress toward carbon neutrality by 2050.

The scope of this study deliberately excluded the examination of legal dimensions of environmental laws or the European Union regulations and directives presently being enacted and incorporated into Maltese legislation. Moreover, this study did not analyze the role of the Environment and Resources Authority in fulfilling the objectives associated with environmental standards, including the International Financial Reporting Standards and the European Sustainability Reporting Standards. These areas fell outside the intended focus of this research.

A potential source of bias in this study might arise from the methodology of the questionnaire, in which concerns might be raised regarding the honesty of responses and the level of effort participants invested in completing the questions. It is possible that respondents may rush through the questionnaire, thus increasing the risk of inaccurate or misleading answers.

Future research could build on the findings of this study by examining additional organizational and contextual factors that may influence the awareness and adoption of EMA, including organizational culture, management support, regulatory pressure, and business uncertainty. Furthermore, given the multidisciplinary nature of EMA, extending the investigation to other professional groups, such as engineers, environmental specialists, and internal auditors, would provide a deep understanding of the drivers and barriers to EMA implementation. Finally, qualitative case studies focusing on organizations that have successfully integrated EMA, particularly large entities and leading accounting firms, could provide valuable insights into best practices and implementation strategies. These studies would further support the development of more effective sustainability accounting practices within Malta and comparable small island economies.

Author Contributions

Conceptualization, D.T. and F.D.; methodology, D.T., F.D., P.J.B., and S.G.; formal analysis, D.T. and F.D.; investigation, D.T. and F.D.; writing—original draft, D.T. and F.D.; writing—review and editing, F.D., P.J.B., and S.G.; supervision, F.D., P.J.B., and S.G. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

Informed consent was obtained from all subjects involved in the study.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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